

Module 3: Syntactic Parsing & Grammars

CFG, Chomsky Normal Form (CNF), CYK Dynamic Parsing Trace & Dependency Transition Systems

Module III: Semantics & Lexical Vector Resources — Complete 11-Topic Tracker

Topic 25: Lexical Semantics (Synonymy, Antonymy, Hyponymy, Meronymy)

Topic 27 & 28: WordNet Lexical Database (Synsets & Hypernym Hierarchies)

Topic 30: Cosine Similarity Formulation in Vector Spaces

Topic 32: TF-IDF Weighting Formulation & Numerical Analysis [UPLOADED PYQ]

Topic 34: GloVe (Global Vectors from Word-Word Co-Occurrence Counts)

Topic 26: Word Sense Disambiguation (Simplified Lesk & Supervised WSD)

Topic 29: Semantic Similarity (Shortest Path Length in WordNet)

Topic 31: Vector Space Models (Term-Document & Term-Term Matrices)

Topic 33: Word2Vec Embeddings (CBOW vs. Skip-Gram Architectures) [UPLOADED PYQ]

Topic 35: FastText (Subword Character N-Gram Vector Representations)

Topic 25 – 29: Lexical Semantics, WordNet & WSD Algorithms

Lexical Relation	Formal Definition	Representative Example
1. Synonymy	Different words expressing the identical or near-identical concept ($A \approx B$).	"large" $\leftrightarrow$ "big", "automobile" $\leftrightarrow$ "car".
2. Antonymy	Words expressing opposite semantic meanings along a spectrum ($A \leftrightarrow -A$).	"hot" $\leftrightarrow$ "cold", "increase" $\leftrightarrow$ "decrease".
3. Hyponymy / Hypernymy	IS-A class-subclass hierarchical relationship (A is a subtype of B).	"sparrow" is a hyponym of "bird"; "animal" is a hypernym of "dog".
4. Meronymy / Holonymy	PART-OF compositional relationship (A is a physical component of B).	"wheel" is a meronym of "car"; "car" is a holonym of "wheel".

The Simplified Lesk WSD Algorithm (Lesk, 1986)

Given target ambiguous word w in context sentence C :

1. Retrieve all dictionary definitions (glosses) and usage examples for each synset sense S_i of w in WordNet.
2. Compute the word token overlap score between the context sentence C and the gloss/examples of sense S_i :

$$\text{Overlap}(S_i, C) = |\text{Words}(S_i.\text{gloss}) \cap \text{Words}(C)|.$$
3. Select the sense $\hat{S} = \arg \max_{S_i} \text{Overlap}(S_i, C)$ with the highest word overlap count.

Topic 30 – 32: Vector Space Models & TF-IDF Weighting [UPLOADED PYQ]

1. Term Frequency-Inverse Document Frequency (TF-IDF) [UPLOADED PYQ]:

$$\text{TF}(t, d) = \frac{\text{Count}(t, d)}{\sum_{t' \in d} \text{Count}(t', d)}$$

$$\text{IDF}(t, D) = \log_{10} \left(\frac{N}{\text{DF}(t)} \right)$$

$$\text{TF-IDF}(\mathbf{t}, \mathbf{d}, \mathbf{D}) = \text{TF}(\mathbf{t}, \mathbf{d}) \times \text{IDF}(\mathbf{t}, \mathbf{D})$$

Where N is total documents in corpus, and $\text{DF}(t)$ is number of documents containing term t .

2. Vector Cosine Similarity Metric:

$$\cos(\theta) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2} = \frac{\sum_{i=1}^d u_i v_i}{\sqrt{\sum_{i=1}^d u_i^2} \sqrt{\sum_{i=1}^d v_i^2}}$$

Topic 33 – 35: Neural Word Embeddings (Word2Vec, GloVe, FastText) [UPLOADED PYQ]

The Distributional Hypothesis (J.R. Firth, 1957)

"You shall know a word by the company it keeps!" Words appearing in similar contextual distributions acquire geometrically close spatial representations in d -dimensional embedding space.

Embedding Model	Training Objective & Operational Mechanism	Key Advantages & Tradeoffs
1. Word2Vec: CBOW [UPLOADED PYQ]	Continuous Bag-of-Words: Predicts current center target word w_t given surrounding context window words: $\arg \max \log P(w_t w_{t-c}, \dots, w_{t+c})$.	Fast to train; higher accuracy for frequent words; smooths over distributional noise.
2. Word2Vec: Skip-Gram [UPLOADED PYQ]	Predicts surrounding context words given center word w_t : $\arg \max \sum \log P(w_{t+j} w_t)$ using Negative Sampling.	Superb for capturing subtle semantic nuances of rare words; slower to train.
3. GloVe	Global Vectors (Stanford): Factorizes global log-co-occurrence matrix: $J = \sum f(X_{ij})(w_i^T \tilde{w}_j + b_i + \tilde{b}_j - \log X_{ij})^2$.	Combines global matrix factorization statistics with local context window advantages.
4. FastText (Meta AI)	Represents each word as a bag of Character n-Grams (e.g., 'n', 'nn', 'nnn').	Seamlessly computes embeddings for Out-of-Vocabulary (OOV) and morphologically complex words!

M3 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Explain the Continuous Bag of Words (CBOW) and Skip-Gram models of Word2Vec with neat network diagrams. (10 Marks)

- **CBOW Architecture:** Input layer takes $2C$ context word one-hot vectors $\rightarrow$ Projection layer averages context embeddings $\rightarrow$ Softmax output layer predicts the single probability distribution of center target word w_t .
- **Skip-Gram Architecture:** Input layer takes 1 center word one-hot vector $w_t \rightarrow$ Projection layer retrieves embedding $\rightarrow$ Output layer simultaneously predicts $2C$ context word probability distributions $\langle w_{t-C}, \dots, w_{t+C} \rangle$ using Negative Sampling ($\log \sigma(v'_{w_o}{}^T v_{w_I}) + \sum_{i=1}^k \mathbb{E}[\log \sigma(-v'_{w_i}{}^T v_{w_I})]$).